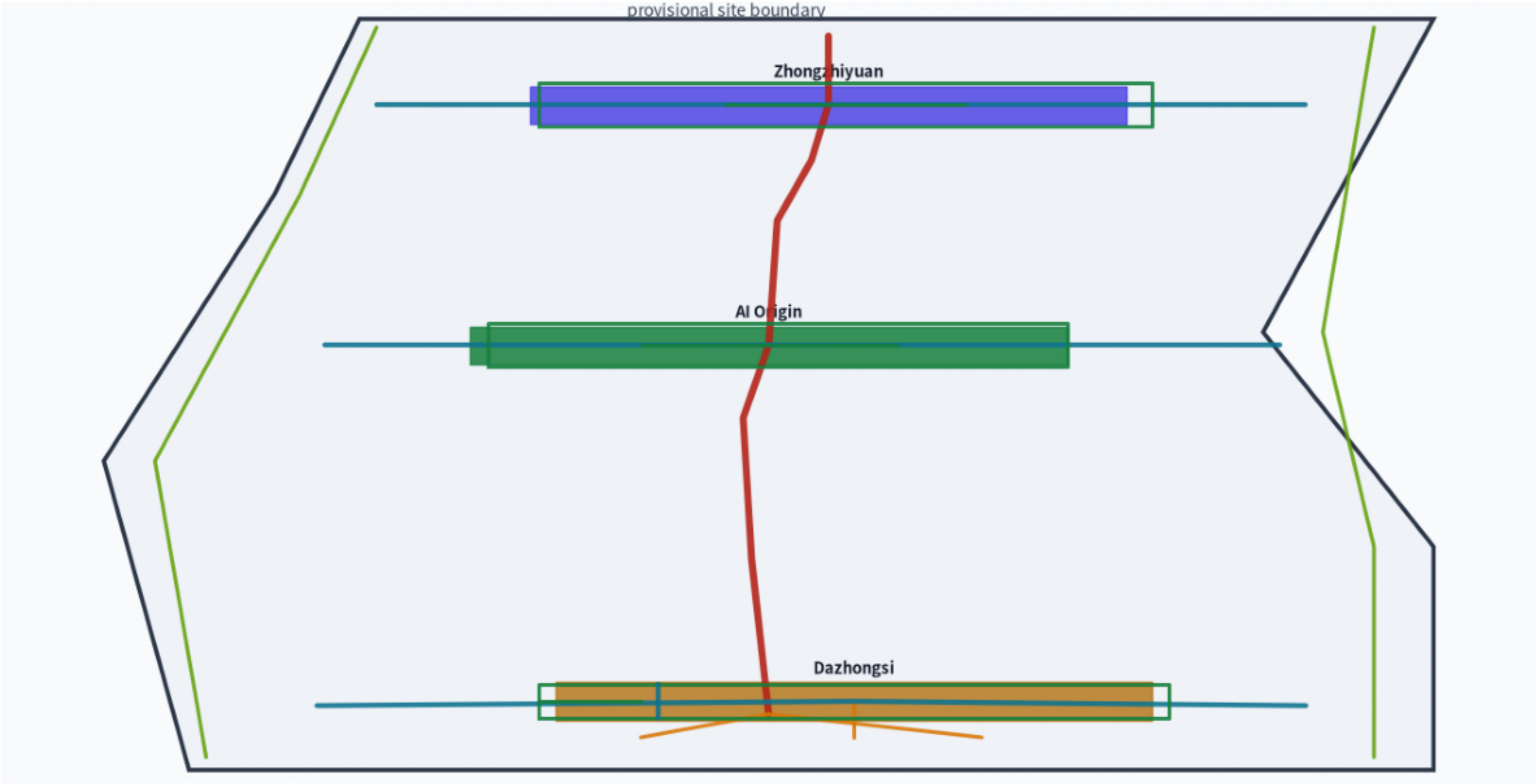




Evidence chain and package



19.20      2.802      8813 m      17/18      80.34      69.1%      25.2%

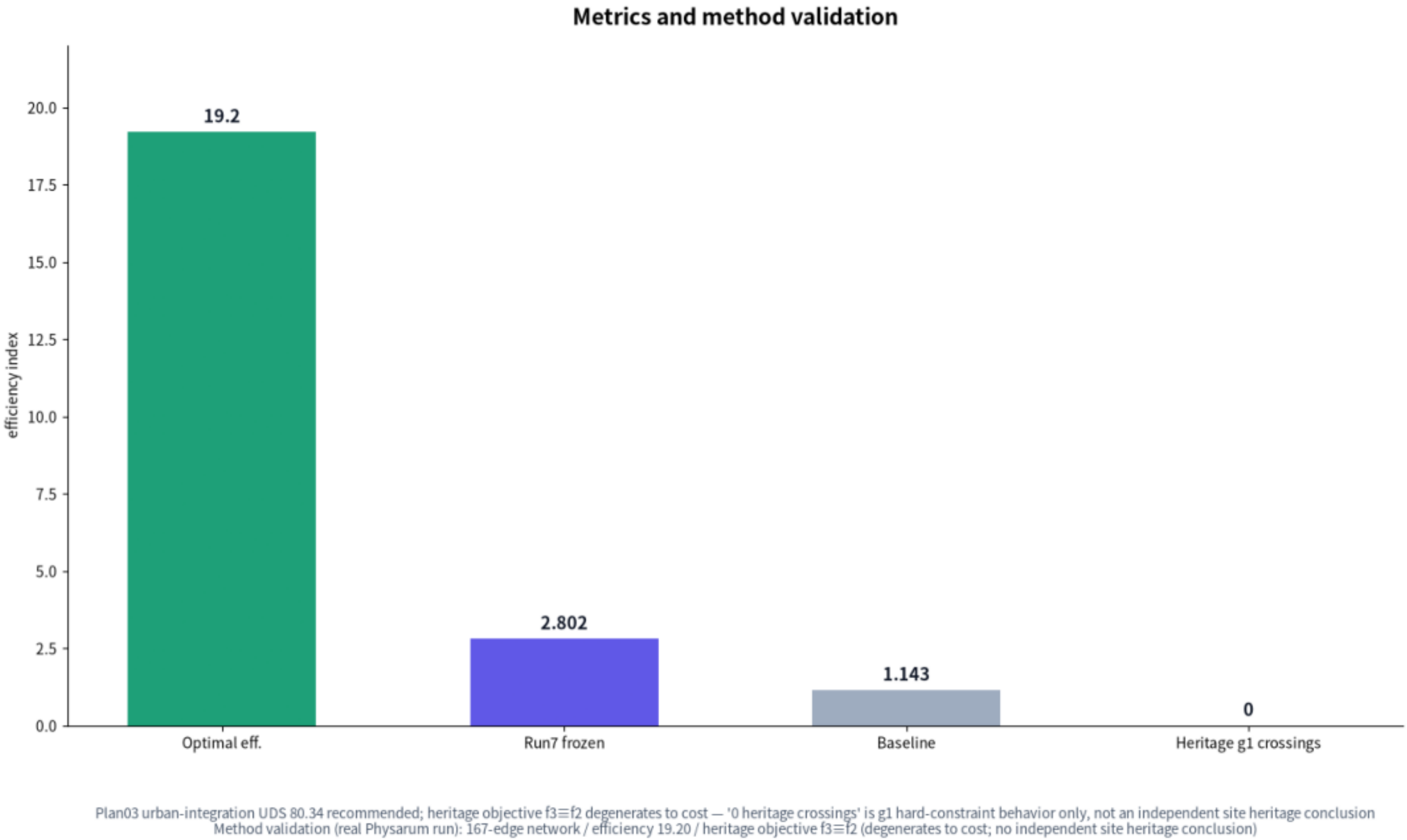
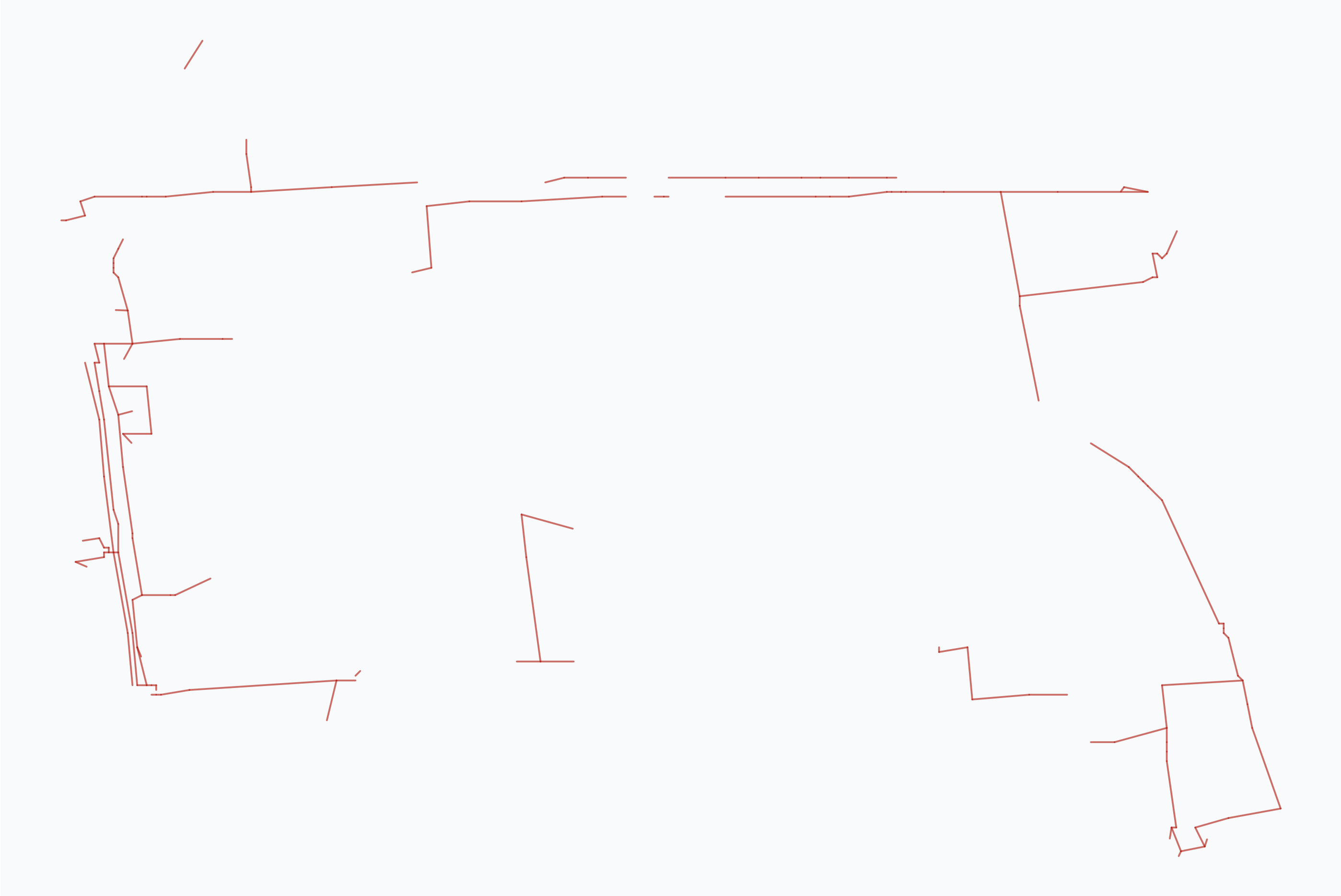
最优效率 optimal eff.      Run7 冻结 frozen      骨架总长 skeleton      关键节点覆盖 cover      Plan03 UDS      透水铺装率 permeable      绿色渗透率 green pen.

Method: Bio-Physarum adaptive network (Tero et al. 2010) + NSGA-II multi-objective | Method validation (real Physarum run): 167-edge network / efficiency 19.20 / heritage objective f3≡f2 (degenerates to cost; no independent site heritage conclusion)

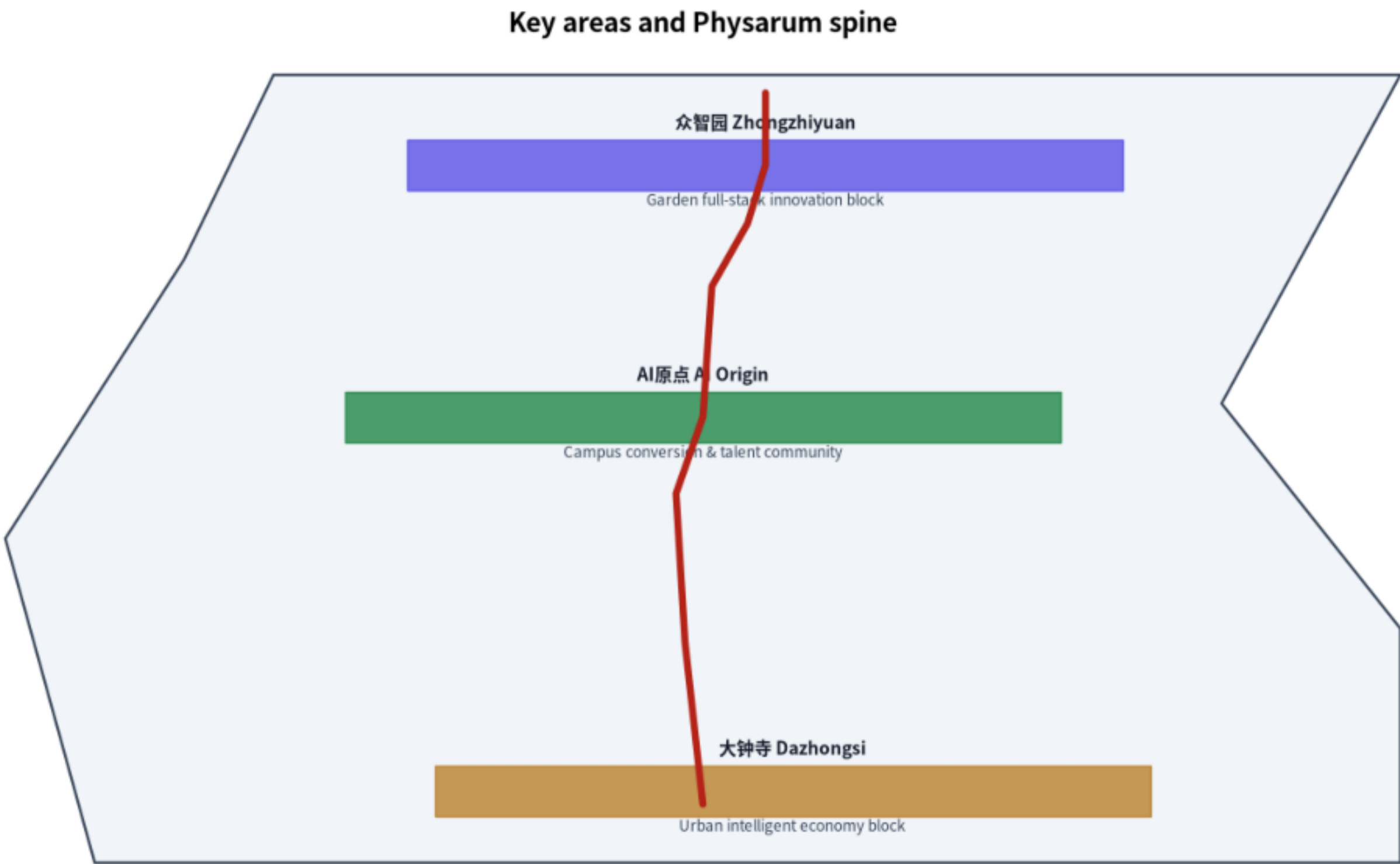


Method: Bio-Physarum + NSGA-II

Real Physarum 167-edge skeleton (method validation)







Method: Bio-Physarum adaptive network (Tero et al. 2010) + NSGA-II multi-objective

Global Benchmark Cases

| 案例                     | 地点          | 核心做法（公开报道）                         | 借鉴/警示                    |
|------------------------|-------------|------------------------------------|--------------------------|
| Sidewalk Labs Quayside | Toronto     | Sensing, data trust (2017-2020)    | Governance / caution     |
| Masdar City            | Abu Dhabi   | Low-emission, passive cooling, PRT | Low-carbon + feeder      |
| Songdo IBD             | Incheon     | Reclaimed town, U-City sensing     | New infra reserve        |
| Amsterdam              | Netherlands | PPP, living-lab                    | Incremental + governance |
| Xiong'an               | Hebei       | CIM twin, blue-green, tunnels      | Twin + blue-green        |



Test Scenarios & Implementation Matrix

Test Validation Scenarios

| 场景                      | 验证对象                        | 检验依据                       | 关键目标                     |
|-------------------------|-----------------------------|----------------------------|--------------------------|
| T-01 Dazhongsi TOD      | Four-quadrant link & feeder | GB 50763-2012 / GB/T 51328 | Barrier-free flow        |
| T-02 Xiaoyue stormwater | Blue-green + permeable      | Sponge guide / GB 50014    | Permeable 69.1% / 25.2%  |
| T-03 AI Origin access   | Campus-park stitching       | GB 50763-2012              | Loop covers + continuous |

Implementation Matrix (suggestive)

| 项目                      | 建议责任主体              | 建议周期 | 暂停/退出                  |
|-------------------------|---------------------|------|------------------------|
| JZ-01 Slow stitching    | Transport + park    | Near | Pause if unsettled     |
| JZ-02 Qinghe interface  | Water + park        | Mid  | Exit if flood unmet    |
| JZ-03 Conversion street | University + bureau | Mid  | Pause if dispute       |
| JZ-04 Four-quadrant     | Rail + transport    | Mid  | Postpone if mismatch   |
| JZ-05 Edge computing    | Bureau + operator   | Long | Exit if absent         |
| JZ-06 Recalibration     | Design + university | Long | Keep 'to be confirmed' |



Accessibility Matrix (GB 50763-2012)

| 画像                    | 无障碍与包容性需求                        | 空间响应                                   |
|-----------------------|----------------------------------|----------------------------------------|
| Open-source dev       | Night collaboration safety       | Barrier-free entrance/toilet, lighting |
| Startup team          | Members with disabilities        | Barrier-free office/test space         |
| Enterprise visitor    | International / reduced mobility | Multilingual signage, elevator         |
| Nearby resident       | Elderly/children/disabled        | Curb ramps, tactile, seats             |
| Univ. teacher/student | Disabled connectivity            | Barrier-free stitching, point          |

Risk, Copyright & Compliance

• Real Physarum lies ~2-3 km west of provisional boundary (~140 m overlap); no translation, no fabricated redline.

• Heritage protection is editable\_by\_agent=false; official geometry missing, f3≡f2 (heritage degenerates to cost).

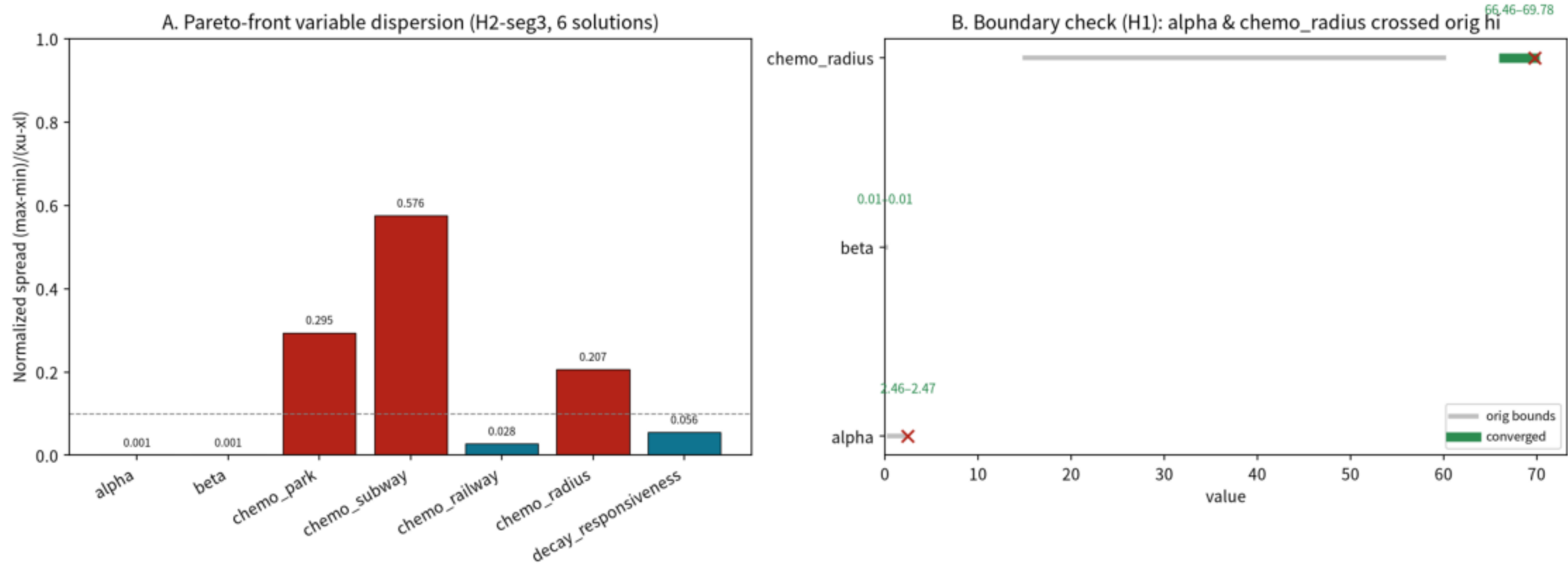
• No claim of official approval, approved plan, final ownership, or guaranteed implementation.

• HTML loads no remote scripts/tiles/fonts/iframes/forms/APIs.

• Appeal/redress: discover-register-rectify-revisit loop (suggestive).



Parameter sensitivity — Bio-Physarum + NSGA-II (real runs, method validation)



Cost Breakdown (suggestive)

| 等级        | 长度(m)  | 单价(元/m) | 造价(万元) |
|-----------|--------|---------|--------|
| Arterial  | 740.2  | 8000    | 592.2  |
| Secondary | 1744.1 | 5000    | 872.0  |
| Branch    | 3679.9 | 3000    | 1104.0 |
| Micro     | 2648.8 | 1500    | 397.3  |
| Total     | 8813.0 | —       | 2965.5 |

Avg 3365 CNY/m; municipal road only, excl. utility/rail/demo (suggestive)

Policy Alignment (suggestive)

| 项目    | 政策/标准依据 (真实)                     | 待确认         |
|-------|----------------------------------|-------------|
| JZ-01 | GB 50763-2012 / barrier-free reg | ownership   |
| JZ-02 | Sponge guide / GB 50014          | blue line   |
| JZ-03 | Beijing renewal reg (suggestive) | clauses     |
| JZ-04 | GB/T 51328 / GB 50763            | station     |
| JZ-05 | New-infra policy (to confirm)    | regulator   |
| JZ-06 | 京政发〔1984〕128号                    | map/redline |

Suggestive phasing gantt (6 projects x 3 phases)

